CARETAG: STANDALONE INTELLIGENT SAFETY MONITORING SYSTEM

Mid-Semester Evaluation | CPG No. 330

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Academic Context

Programme: BE Fourth Year, CoE / CoSE

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PROBLEM STATEMENT

Static Geofencing is Reactive & Context-Blind

Scale of the Challenge

1.5B+

Elderly by 2050

People aged 60+ globally; wandering affects 17–40% of dementia cases

8.8M

Indians with Dementia

A population acutely vulnerable to unsafe wandering episodes

1.48L+

Crimes vs. Children

NCRB-recorded crimes against children in India (2019)

Three Structural Failures of Existing Systems

Platforms such as Family Link, Life360, and Find My share a common architectural flaw — they reduce safety to a single binary question: *is the user inside or outside the boundary?*

No Routine Awareness

Systems have no model of the individual's habitual movement patterns.

No Temporal Context

Alerts carry no time-of-day awareness — a midnight deviation looks identical to a noon one.

Binary Alerts Only

No severity measure; every boundary crossing triggers the same alarm regardless of risk level.

LITERATURE REVIEW

Research Gaps & Problem Definition

A systematic review of literature and commercial products identified **five structural gaps** that no existing solution addresses in combination.

1

Static Geofences Cannot Model Routines

Li et al. (2008) — individual behavioral patterns are not encoded in any current geofencing product.

2

No Temporal Context in Alerts

Liao et al. (2007); Zheng et al. (2009) — time-of-day is a critical predictor of expected location, yet is universally ignored.

3

Binary Alerting — No Severity Signal

Ji et al. (2020) — a continuous anomaly score is far more actionable than a simple in/out flag.

4

Personalized Geofences Not Auto-Constructed

Lin et al. (2018) — caregivers must manually draw zones; trajectory mining can automate this.

5

Real-Time Scoring Requires No-Retrain Architecture

Mazimpaka & Timpf (2016) — most systems require batch retraining; live scoring per GPS point is unsolved.

Problem Statement: GPS safety systems today cannot distinguish routine-but-unregistered visits from genuine risk, and provide no quantified anomaly measure to guide caregiver response.

PROJECT SCOPE

Approved Objectives & Scope

Six objectives were formally approved at the proposal evaluation stage. Each maps directly to a pipeline stage or system component.

1 Stay-Point Extraction & Clustering

Li et al. algorithm applied to GPS history; K-Means clustering produces named locations representing the individual's routine geography.

2 Dual-Input LSTM Location Predictor

Predicts expected cluster by time-of-day using both temporal slot and recent movement history as inputs.

3 Real-Time Anomaly Scorer

Weighted spatial + temporal deviation score in [0, 1]; anomaly flagged when score exceeds threshold of **0.6**.

4 React + Leaflet Caregiver Dashboard

Interactive geofence management and live simulated tracking interface for caregivers.

5 Multi-Phase GPS Simulation Engine

Generates synthetic GPS streams across STAY, COMMUTE, and ANOMALY phases with realistic jitter and telemetry.

6 Offline-First Architecture with Optional Backend

localStorage-based standalone mode; optional .NET 8 + PostgreSQL + FastAPI + SignalR full-stack deployment.

In Scope

- ML pipeline (extraction → clustering → LSTM → scoring)
- Caregiver dashboard & simulation engine
- Offline-first + optional backend modes

Out of Scope

- Physical hardware & live field deployment
- Multi-user authentication
- Native mobile application

LITERATURE SURVEY

Prior Work & CareTag's Differentiation

Study	Contribution	CareTag's Extension
Li et al. 2008	Stay-point detection algorithm	Used directly as Stage 1 of the ML pipeline
Liao et al. 2007	Temporal context improves prediction accuracy	Encoded as time-slot input to the dual-input LSTM
Zheng et al. 2009	Sequential visit patterns predict next location	Basis for LSTM sequence modeling architecture
Ji et al. 2020	Spatial + semantic signals outperform spatial-only	Both signals combined into a single deviation score
Lin et al. 2018	Trajectory-mined personalized geofences	Cluster centroids used as geofence anchors
Mazimpaka & Timpf 2016	Most systems require batch retraining	CareTag scores each GPS point live — no retrain needed

Net Differentiator: Existing work solves clustering *or* sequence modeling *or* geofencing in isolation. CareTag is the first to connect all three into one live, offline-capable caregiver tool.

SYSTEM OVERVIEW

System Overview & Core Design Philosophy

ML Engine

Learns the individual's behavioral routine from GPS history using the **Microsoft GeoLife dataset** — 182 users, 17,621 trajectories.

- Stay-point extraction from raw .plt files
- K-Means clustering into named routine locations
- Dual-input LSTM for time-aware location prediction
- Live anomaly scorer outputting a continuous [0,1] score

Geofence Dashboard

A React + Leaflet caregiver interface providing real-time visibility into the monitored individual's location and behavioral state.

- Draw, name, and delete geofence zones on an OSM map
- Watch live simulated GPS tracking with telemetry
- Receive and dismiss anomaly alert banners
- Operates fully offline or with Dockerized backend

Core Design Principle: Treat safety monitoring as a *behavioral deviation problem*, not a boundary-crossing problem. The system asks not "where is the user?" but "is this location unusual given who they are and what time it is?"



ARCHITECTURE

System Architecture — 3-Tier Design

1

Tier 1 — Data Layer

Current: Microsoft GeoLife v1.3 .plt trajectory files (182 users, 17,621 trajectories)
Planned: ESP32 microcontroller + Neo-6M GPS module via GSM network

2

Tier 2 — ML Engine

Stay-point extraction → K-Means clustering → Dual-input LSTM → Anomaly scorer
Artifacts persisted as `.joblib` (scaler/cluster model) and `.pth` (LSTM weights)

3

Tier 3 — Application Layer

React 18 dashboard ↔ .NET 8 Minimal REST API ↔ PostgreSQL 15.x
FastAPI anomaly scoring endpoint · SignalR real-time push to browser

Standalone Mode

Browser-only operation using `localStorage` for geofence persistence. No backend required. Simulation engine runs entirely client-side.

Full-Stack Mode

Dockerized backend with .NET 8 API, FastAPI scorer, PostgreSQL database, and SignalR hub for real-time push notifications to the dashboard.

UML DESIGN

UML Design Diagrams

Use Case — Caregiver Actor

Geofence Management

Create geofence · Name geofence · Delete geofence

Simulation & Monitoring

Start simulation · View live telemetry · Advance GPS step

Alert Handling

Receive anomaly alert · Dismiss alert banner

Sequence — Anomaly Alert Flow

01

Advance Step

Caregiver triggers next GPS point in simulation

02

POST /simulate/step

Dashboard sends GPS coordinate to FastAPI scoring endpoint

03

LSTM + Haversine Scoring

FastAPI computes spatial and temporal deviation scores

04

Threshold Check

If combined score > 0.6 → SignalR `AlertFired` event emitted

05

Dashboard Renders Alert

Dismissible alert banner appears with score and location context

Activity Diagram Logic: Every incoming GPS point runs two parallel checks simultaneously — geofence point-in-polygon test AND behavioral anomaly scoring — ensuring neither boundary nor behavioral violations are missed.

ML PIPELINE

ML Pipeline — Detailed Design

Stage 1 — Stay-Point Extraction

Algorithm: Li et al. (2008)

- Dwell threshold: ≥ 20 minutes
- Spatial radius: 200 m
- Output: **67 stay points** from 50 trajectory files

Stage 2 — Clustering & Routine Modeling

Algorithm: K-Means (K ≤ 10, elbow method)

- Cluster centroids → named routine locations
- Dual-input LSTM: time-slot + recent history
- Output: predicted expected cluster per time window

Stage 3 — Anomaly Scoring

Formula: deviation = α × spatial_score + (1−α) × temporal_score

- α = 0.5 (equal weighting)
- Spatial = normalized Haversine distance
- Temporal = 1 − LSTM softmax probability
- Flag anomaly if score > **0.6**

The scoring formula produces a continuous value in **[0, 1]** where 0 represents fully expected behavior and 1 represents maximum deviation. The 0.6 threshold was selected to balance sensitivity against false-positive rate during simulation testing.

$$\text{deviation} = \alpha \quad d_{\text{spatial}} + (1 - \alpha) \quad (1 - P_{\text{LSTM}}), \quad \alpha = 0.5$$

DASHBOARD & SIMULATION

Dashboard, Simulation & UI Design

Map Panel

Built on **Leaflet.js + OpenStreetMap tiles**. Caregivers interact directly with the map to:

- Draw geofence polygons by clicking boundary points
- Assign human-readable names to each zone
- Delete zones with a single action
- View live GPS marker updating in real time

Simulation Engine

Generates a synthetic GPS stream to validate the full pipeline without live hardware:

- GPS packet emitted every **3 seconds**
- Three phases: **STAY** → **COMMUTE** → **ANOMALY**
- Realistic position jitter, speed, and battery telemetry

Live Telemetry Sidebar

- Real-time metrics: coordinates, speed, anomaly score
- Dismissible alert banner on threshold breach

Offline Mode

Fully functional via `localStorage`. Geofences and simulation state persist across browser sessions with no backend dependency.

Full-Stack Mode

Switches automatically to PostgreSQL persistence and SignalR real-time push when the Dockerized backend is detected and running.

Dual-Mode Design Rationale

Ensures the system remains demonstrable in any environment — from a laptop with no internet to a fully deployed cloud instance.

TECHNOLOGY STACK

Tools & Technology Stack

Layer	Technology
ML / Data Science	Python · pandas · scikit-learn (K-Means, preprocessing) · PyTorch (LSTM training & inference)
Frontend	React 18 · Leaflet.js · OpenStreetMap tiles
Backend (optional)	.NET 8 Minimal API · FastAPI (Python scoring endpoint) · SignalR (real-time WebSocket push)
Database	PostgreSQL 15.x
Infrastructure	Docker Compose (multi-container orchestration)
Dataset	Microsoft GeoLife v1.3 — 182 users, 17,621 trajectories, 24.2M GPS points

Design Principles Governing Stack Selection

- **Offline-first:** Core functionality requires no network or server
- **Modularity:** ML engine, API, and frontend are independently deployable
- **Reproducibility:** Docker Compose ensures consistent environment across machines
- **Open standards:** OSM tiles, REST, WebSocket — no proprietary lock-in

Evaluation Readiness

All six approved objectives have corresponding implemented components. The simulation engine enables full end-to-end demonstration without physical GPS hardware, satisfying the evaluation panel's requirement for a working prototype at mid-semester review.

Thank You

Questions & Discussion